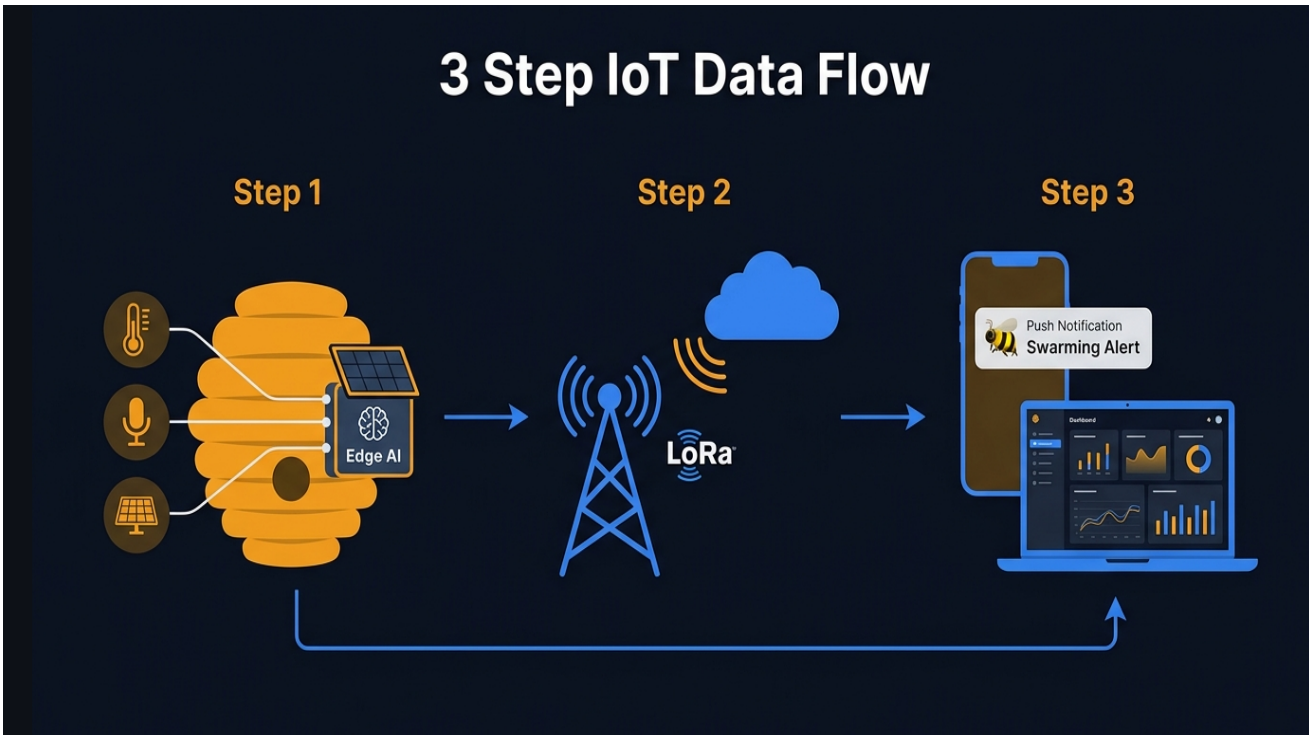
Beevil Knievel Smart Beehive Monitoring System

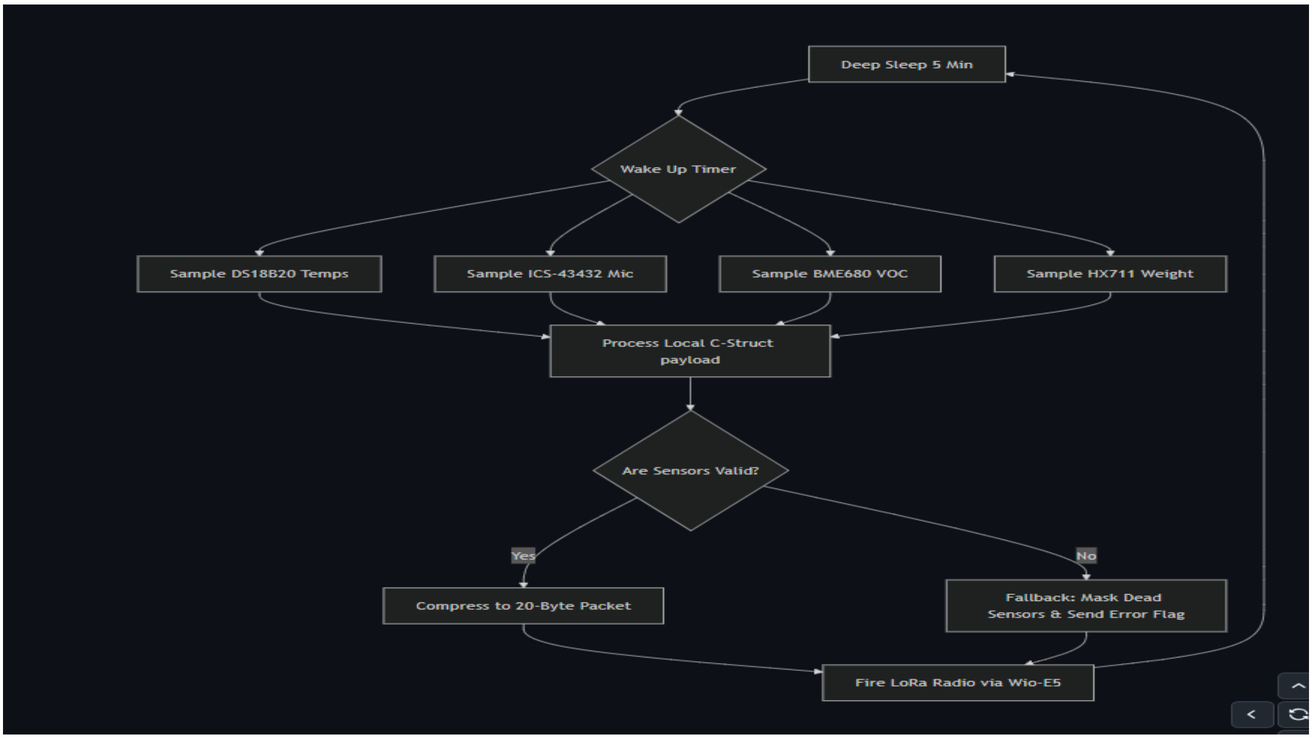
Consolidated Architecture & Firmware Overview

Page 1: End-to-End IoT Data Flow



The hive node periodically wakes from deep sleep, samples DS18B20, ICS-43432 microphone, BME680 environmental sensor and HX711 load-cell. Data is packed into a compact payload and transmitted over LoRa for cloud analytics and farmer notifications.

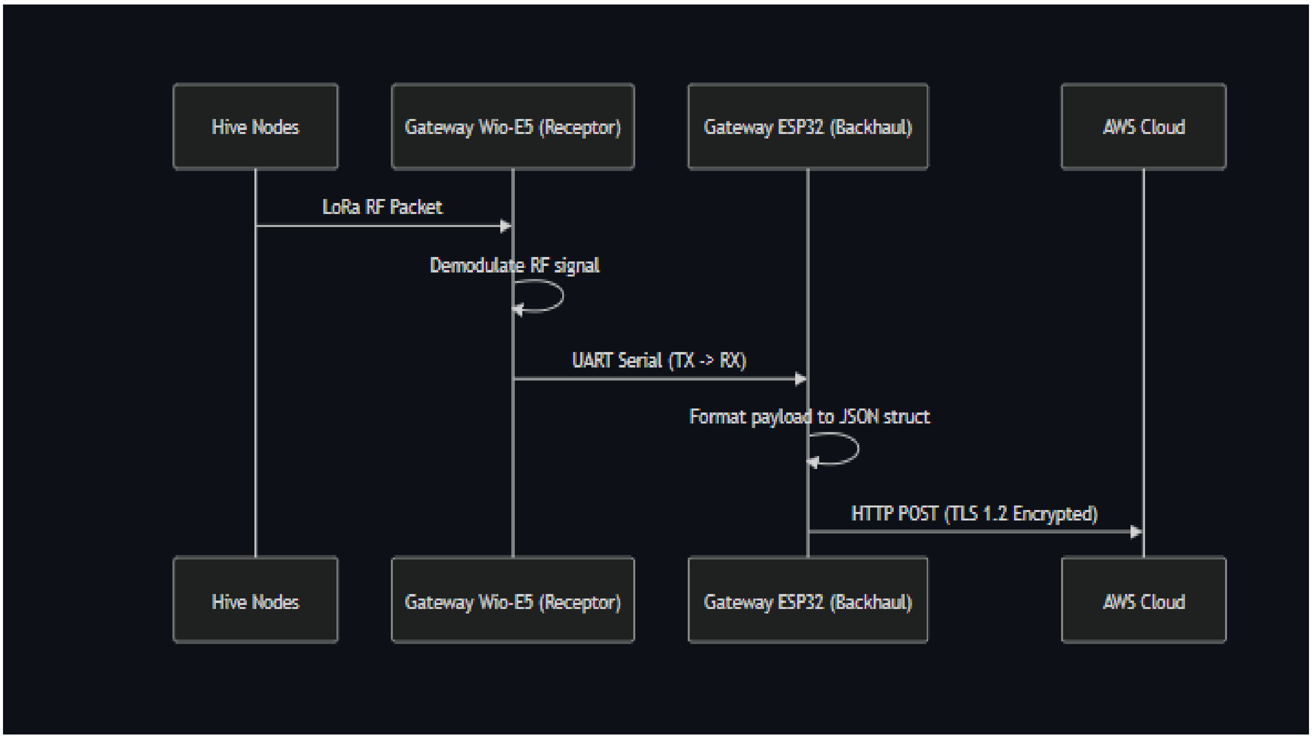
Page 2: Transmitter Firmware



Firmware Features:
- Deep Sleep scheduler (5 min)
- Multi-sensor acquisition
- Sensor validation and fault masking
- Compact C-structure payload
- 20-byte packet compression
- LoRa transmission
- Low-power architecture
- Modular sensor drivers

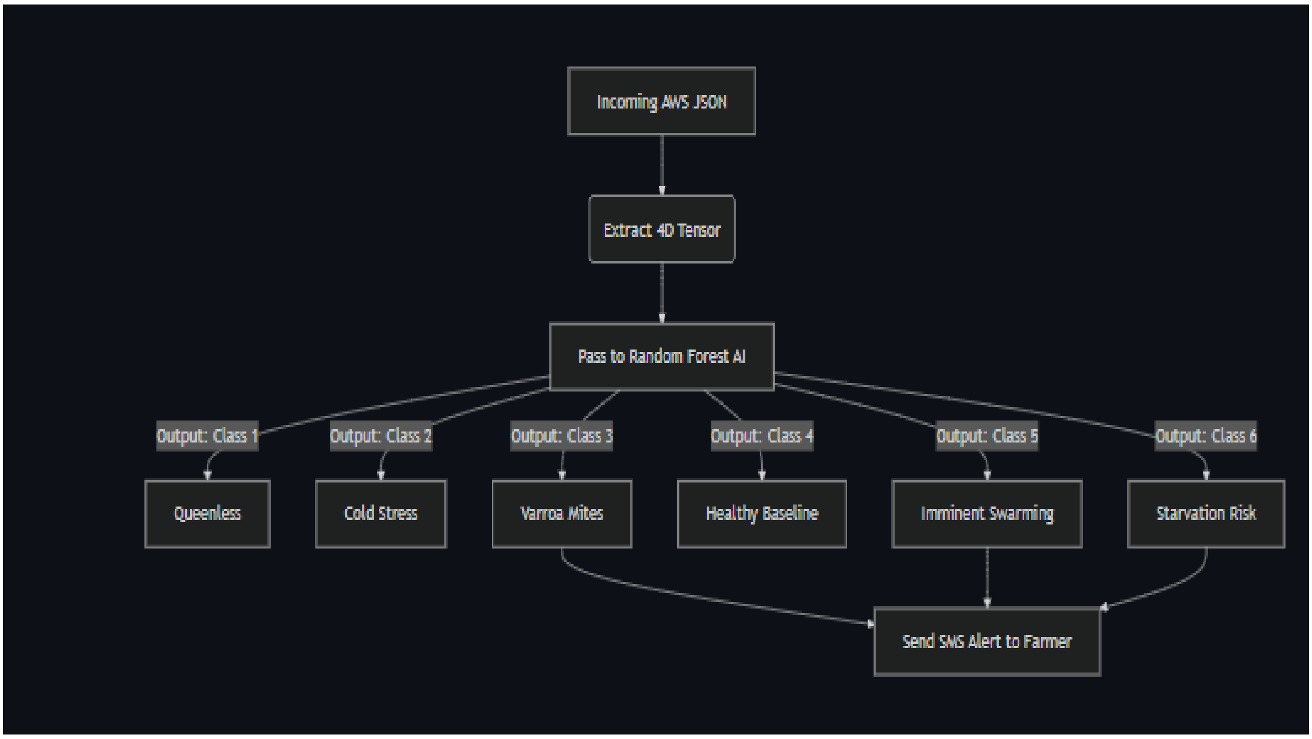
Algorithm:
Boot → Wake Timer → Read Sensors → Validate → Compress → LoRa TX → Sleep

Page 3: Gateway Communication



Gateway Responsibilities:
• Wio-E5 receives LoRa RF packets
• Demodulates packets
• UART forwarding to ESP32
• ESP32 converts payload to JSON
• Secure HTTPS (TLS1.2) upload to AWS
• Retry and buffering support

Page 4: Cloud AI Pipeline



Cloud & AI:
• JSON ingestion
• Tensor extraction
• Random Forest inference
• Six health classes: Healthy, Queenless, Cold Stress, Varroa Mites, Imminent Swarming, Starvation Risk.
• Dashboard visualization
• SMS alerts to farmers.

Component Roles:
ESP32: controller & gateway; Wio-E5: LoRa radio; DS18B20: temperature; ICS-43432: acoustic sensing; BME680: environmental sensing; HX711: hive weight; AWS: storage, analytics, AI, alerts.